Iris Species Classification

Machine Learning · Executive Summary

Dataset	Kaggle – Iris Classification (150 samples, 4 features, 3 classes)
Task	Multi-class supervised classification
Models	K-Nearest Neighbours · Logistic Regression · Decision Tree
Split	80 % train / 20 % test | 5-fold stratified cross-validation
Author	InternSpark Internship Task Submission

1. Dataset & Exploratory Data Analysis

The Iris dataset contains **150 samples** evenly distributed across three species: *Iris-setosa*, *Iris-versicolor*, and *Iris-virginica*. Four numerical features are measured in centimetres: sepal length, sepal width, petal length, and petal width. **No missing values** were detected.

Feature	Min	Mean	Max	Std Dev
Sepal Length	4.30	5.84	7.90	0.83
Sepal Width	2.00	3.05	4.40	0.43
Petal Length	1.00	3.76	6.90	1.76
Petal Width	0.10	1.20	2.50	0.76

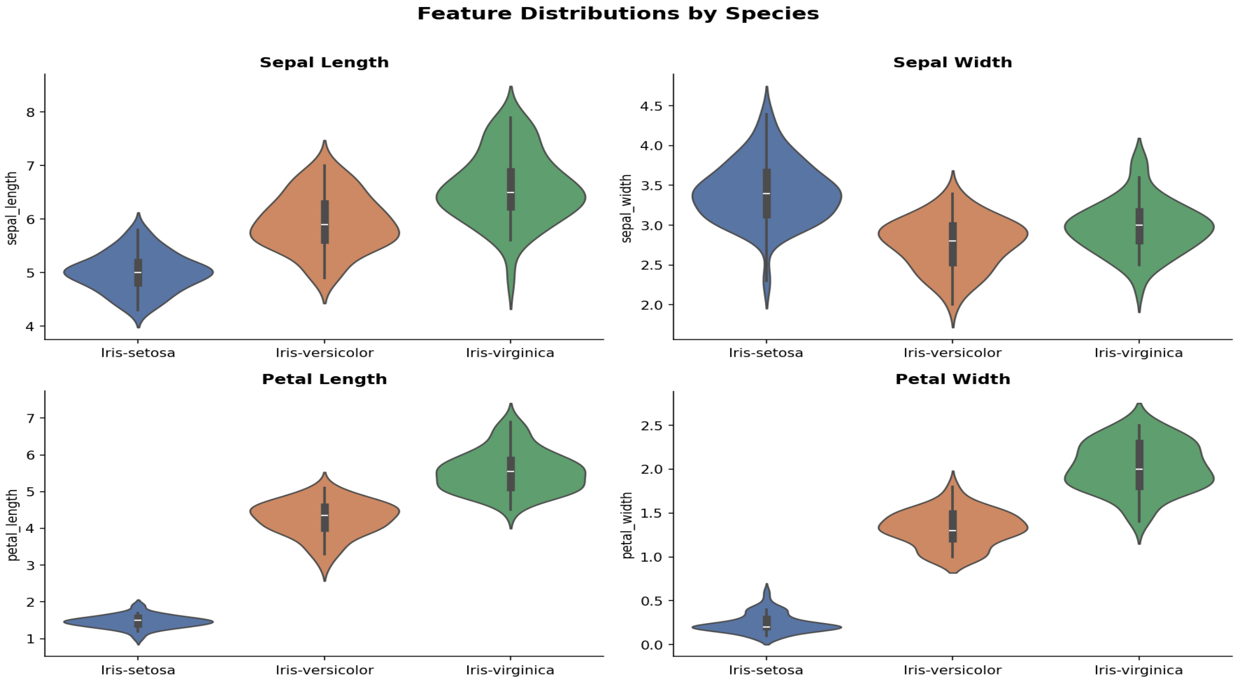

Fig 1. Feature distributions by species (violin plots)

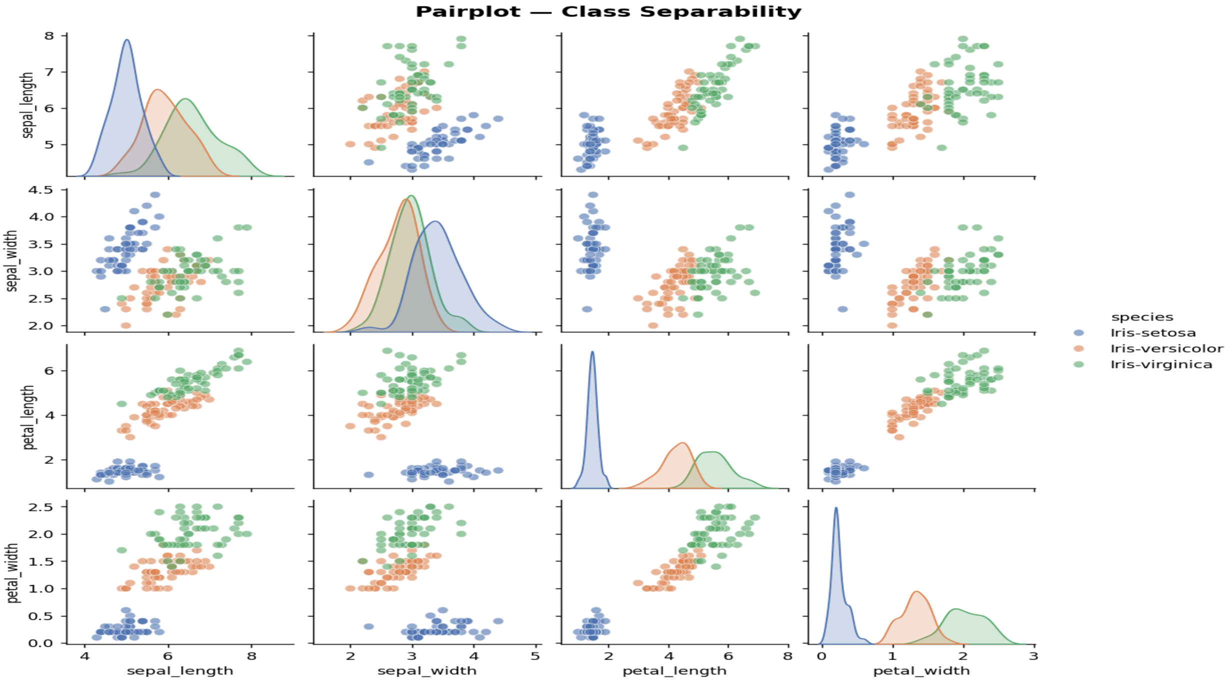

Fig 2. Pairplot showing class separability across all feature pairs

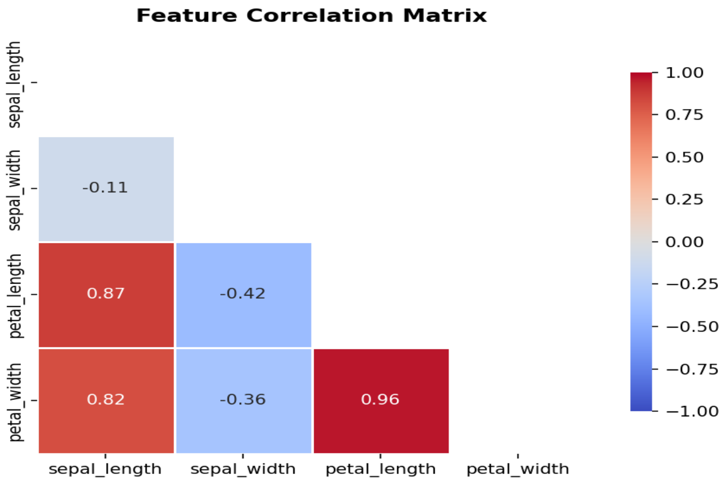

Fig 3. Pearson correlation matrix — petal dimensions highly correlated (r=0.96)

Key EDA Finding: Petal length and petal width provide near-perfect linear separation between *setosa* and the other two classes. *Versicolor* and *virginica* show partial overlap in petal dimensions, making them the harder boundary to learn. Sepal features contribute less discriminative power.

2. Methodology

Step	Details
Pre-processing	StandardScaler (zero mean, unit variance) applied to all features
Train/Test Split	80/20 stratified split (random_state=42); equal class representation
Cross-validation	5-fold StratifiedKFold on training set for unbiased model selection
Hyper-parameters	KNN: k=5 | LR: max_iter=1000, L2 reg | DT: max_depth=4

3. Model Results & Metrics

Model	Test Accuracy	CV Accuracy	CV Std Dev	Recommended
K-Nearest Neighbors (k=5)	93.3 %	95.8 %	± 2.6 %	★ BEST
Logistic Regression	93.3 %	95.8 %	± 2.6 %	
Decision Tree (depth=4)	90.0 %	94.2 %	± 2.0 %	

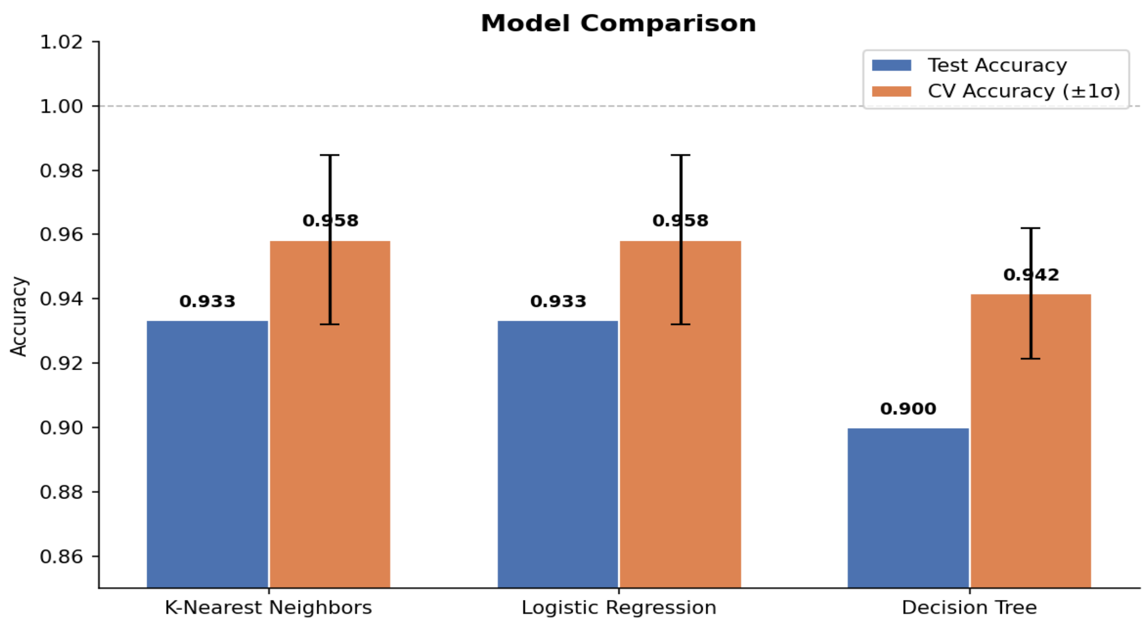

Fig 4. Test vs Cross-validation accuracy comparison across all three models

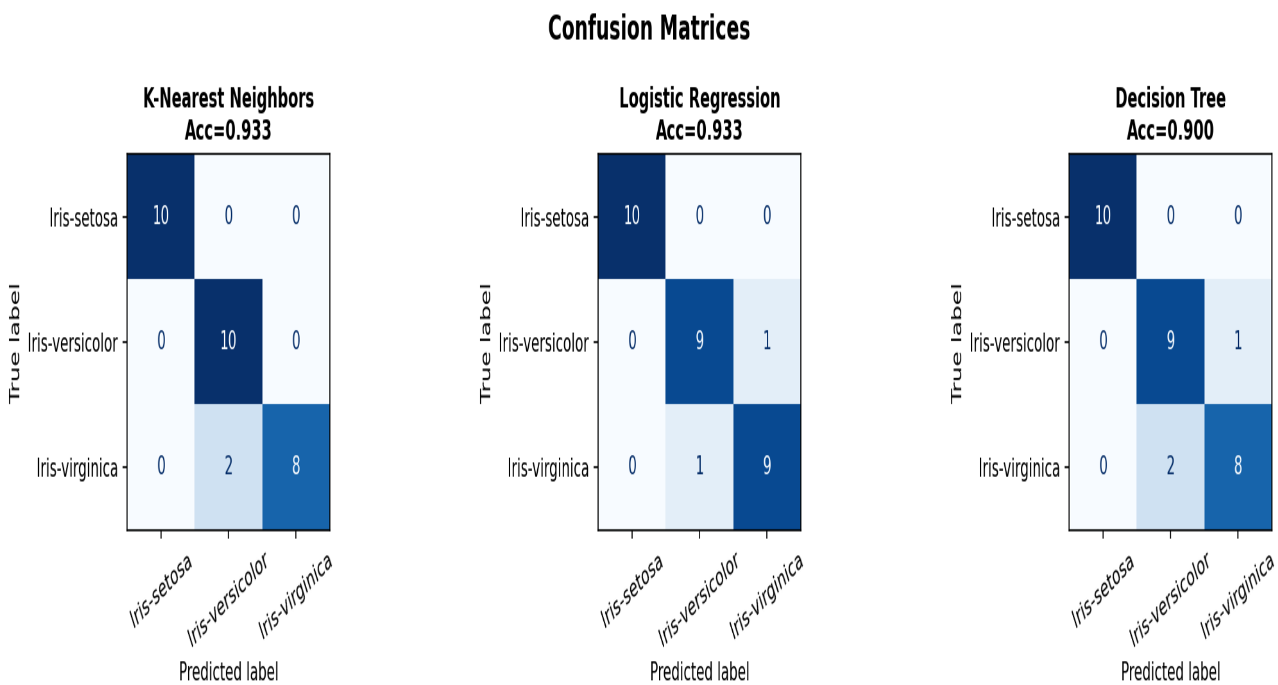

Fig 5. Confusion matrices — all models achieve perfect Setosa classification

Best Model (KNN) — Per-class Metrics:

Class	Precision	Recall	F1-Score	Support
Iris-setosa	1.00	1.00	1.00	10
Iris-versicolor	0.83	1.00	0.91	10
Iris-virginica	1.00	0.80	0.89	10
Macro Average	**0.94**	**0.93**	**0.93**	**30**

4. Decision Tree Interpretability

The Decision Tree (max_depth=4) provides the most interpretable model. Its root split on **petal width ≤ 0.8 cm** perfectly isolates all 50 setosa samples. Subsequent splits use petal length and petal width to separate versicolor from virginica.

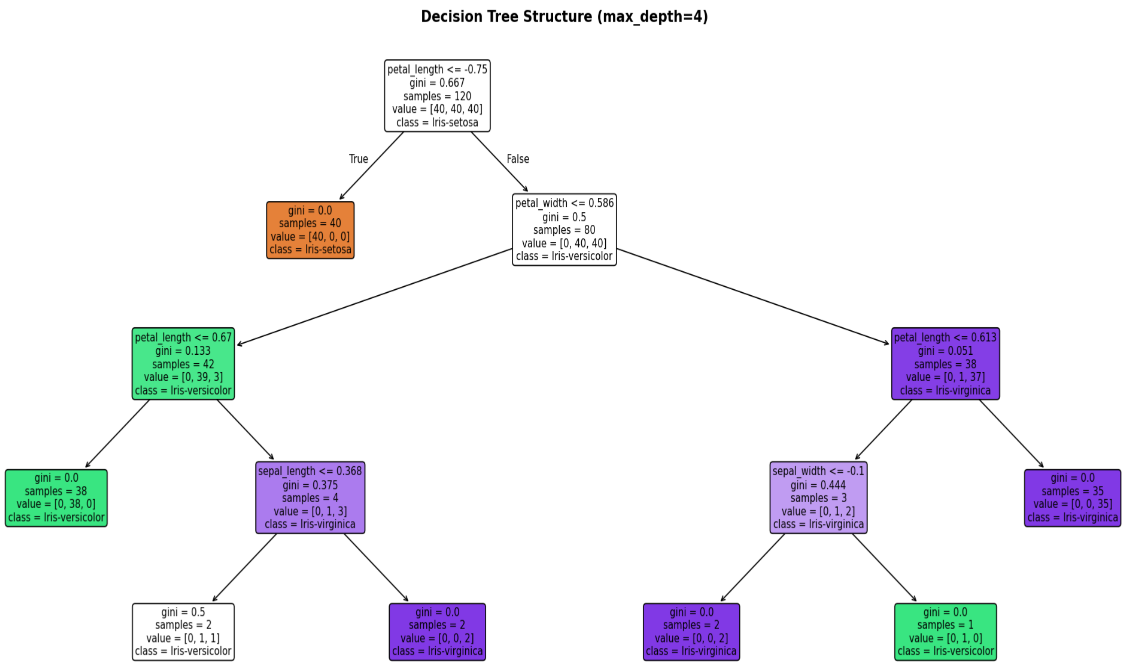

Fig 6. Full Decision Tree structure — petal dimensions dominate all decision boundaries

5. Conclusions & Recommendations

Finding	Detail
Best Model	K-Nearest Neighbours (k=5) — 95.8 % CV accuracy, saved as iris_best_model.joblib
Easiest class	Iris-setosa is linearly separable from other classes (perfect precision & recall on all models)
Hardest boundary	Versicolor vs. Virginica — small overlap in petal dimensions causes the 2-3 misclassifications
Feature importance	Petal length and petal width are the two most discriminative features (r=0.96 between them)

Deployment	Run inference.py with four float arguments for real-time species prediction

All code, the saved model bundle, and plots are available in the project repository. See **README.md** for installation and inference instructions.